OSPG Project Plan — 5-Person Task Breakdown

Oct 31, 2026 · @Anamay

Scope

Five people, three workstreams, one week: replicate Deep Recurrent Optimal Stopping (Damera Venkata & Bhattacharyya, NeurIPS 2023), extend it to American options under rough volatility, and backtest it across equities, FX, commodities and crypto. Seven days is tight, so every workstream runs a reduced version of the full plan.

1. **Replication** — implement OSPG and reproduce the paper's Bermudan max-call, American geometric-call and fractional Brownian motion results against Longstaff–Schwartz, Deep Optimal Stopping and fitted Q-iteration.
2. **Rough-volatility extension** — price American puts under the rough Bergomi model, where the price alone is non-Markovian, priced by the learned policy and compared with published reference prices.
3. **Multi-market backtest** — walk-forward test of learned exit timing on real price windows, comparing a history-aware GRU policy with a memoryless MLP policy and simple rules, across asset classes with different volatility persistence.

The stochastic calculus spine runs through all three: Snell envelope and free boundary (Shreve II, Ch 8), optional stopping, Feynman–Kac benchmarks, and why fBm is not a semimartingale.

Roles at a glance

Each person owns one layer of the codebase and one report section, so no two people write the same file.

Person	Role	Owns in code	Main deliverables
P1	OSPG core and replication lead	`ospg/`, GBM simulator	OSPG loss + GRU/MLP policies; Bermudan max-call and American geometric-call tables

Person	Role	Owns in code	Main deliverables
P2	Baselines and stopping theory	`baselines/`	Longstaff–Schwartz, Deep Optimal Stopping, fitted Q-iteration, binomial/PDE benchmark, baselines run on rough-vol and real data; theory chapter
P3	Stochastic simulation and rough volatility	`simulators/`	Exact fBm and rough Bergomi simulators; fBm table; American put under rough vol
P4	Market data and backtest harness	`data/`, `backtest/`	Clean multi-asset data, windowing, features, walk-forward splits, cost models, rule-based exits
P5	Backtest experiments and statistics	`analysis/`, `experiments/backtest/`	All real-data runs, risk-sensitive reward, bootstrap tests, regime and Hurst analysis, results figures

Fill in names: P1 ___, P2 ___, P3 ___, P4 ___, P5 ___.

Detailed tasks

Every person has a first deliverable others depend on; ship that first, then the experiments.

P1 — OSPG core and replication lead

First deliverable (end of day 2): a working `ospg` package others can call.

- [ ] GBM path simulator (multi-asset, correlated), discounted rewards
- [ ] Policy networks: MLP (2 × 20, batch norm) and GRU (1 × 20), outputting stopping probabilities at every step
- [ ] OSPG loss (Algorithm 1): stopping-time distribution, constant baseline, stop-gradient weights

- [ ] Training loop: Adam, lr 1e-3, early stopping on validation reward; hard-decision evaluation (stop when probability ⩾ 0.5)
- [ ] Replicate Bermudan max-call: d ∈ {20, 50}, s_0 = 100, 3 random splits (full grid only if time allows)
- [ ] Replicate American geometric call (100 exercise dates) and report % error vs the semi-analytic price
- [ ] Support P3 and P5 in plugging their problems into `ospg`
- [ ] Report: Method section (the OSPG derivation in our own notation)

P2 — Baselines and stopping theory

First deliverable (end of day 2): Longstaff–Schwartz and a binomial American put price, so everyone has a ground truth.

- [ ] Binomial tree and finite-difference (Feynman–Kac / free-boundary) pricer for the 1-D American put
- [ ] Longstaff–Schwartz with polynomial basis
- [ ] Deep Optimal Stopping (Becker et al.): one network per time step, backward induction
- [ ] Fitted Q-iteration with MLP and GRU (paper's Appendix C loss)
- [ ] Run Longstaff–Schwartz and Deep Optimal Stopping on P3's rough Bergomi paths and P4's backtest windows
- [ ] Report: Theory chapter — Snell envelope, optional stopping, perpetual and finite American put (Shreve II Ch 8)

P3 — Stochastic simulation and rough volatility

First deliverable (end of day 2): an exact fBm simulator with a covariance check.

- [ ] Exact fBm via Cholesky or Davies–Harte; verify the empirical covariance against 0.5(t^{2H} + s^{2H} − |t − s|^{2H})
- [ ] Replicate the fBm stopping table for H from 0.05 to 0.95; confirm value ≈ 0 at H = 0.5
- [ ] Rough Bergomi simulator (hybrid scheme), validated on European option prices from the literature
- [ ] American put under rough Bergomi: RNN-OSPG vs MLP-OSPG vs Longstaff–Schwartz vs Deep Optimal Stopping, against published reference prices
- [ ] Sweep H (e.g. 0.07, 0.1, 0.3, 0.5) to show the value of memory as H moves from 0.5

- [ ] Report: fBm and rough volatility section, including why fBm is not a semimartingale for $H \neq 1/2$

P4 — Market data and backtest harness

First deliverable (end of day 2): a harness that turns any price series into train/test windows with features and rewards.

- [ ] Loaders: NIFTY 50 and S&P 100 (yfinance), BTC, ETH and top-10 crypto (Binance public data or ccxt); FX and gold only if time allows, adjusted for splits and dividends
- [ ] Cleaning: gaps, outliers, delisted names, stablecoin exclusion; log survivorship choices
- [ ] Windowing: fixed calendar horizon (≈ 1 month, daily bars only)
- [ ] Features from past data only: return so far, drawdown, trailing volatility, lagged returns, market index and VIX
- [ ] Walk-forward splitter with an H-day gap between train, validation and test
- [ ] Cost models per venue (NSE, US, crypto taker fees, perpetual funding)
- [ ] Rule-based exits: hold to end, take-profit/stop-loss, trailing stop
- [ ] Report: Data and backtest methodology section

P5 — Backtest experiments and statistics

First deliverable (end of day 2): the statistics toolkit run on a toy example.

- [ ] Block bootstrap of paired reward differences (resample by date), confidence intervals, multiple-testing correction
- [ ] Memory diagnostics per asset: return and |return| autocorrelation, Hurst exponent of returns and of volatility
- [ ] Run GRU-OSPG, MLP-OSPG, Longstaff–Schwartz and rules across every asset class, walk-forward
- [ ] Two rewards: linear return, and risk-sensitive (exponential utility or mean–variance)
- [ ] Regime slices (2008, 2020, calm years) and cost sensitivity
- [ ] Headline figure: GRU advantage vs each asset's volatility persistence
- [ ] Report: Results and discussion for the backtest; leakage checks

Shared repo and interface

Agree on one data contract in week 1 so every simulator, dataset and method plugs into every other.

The contract: every problem, simulated or real, produces

- `X` — features, shape `[N, H+1, d]` (path i, time step j, feature k), built only from data up to step j
- `R` — rewards, shape `[N, H+1]` (reward for stopping path i at step j)

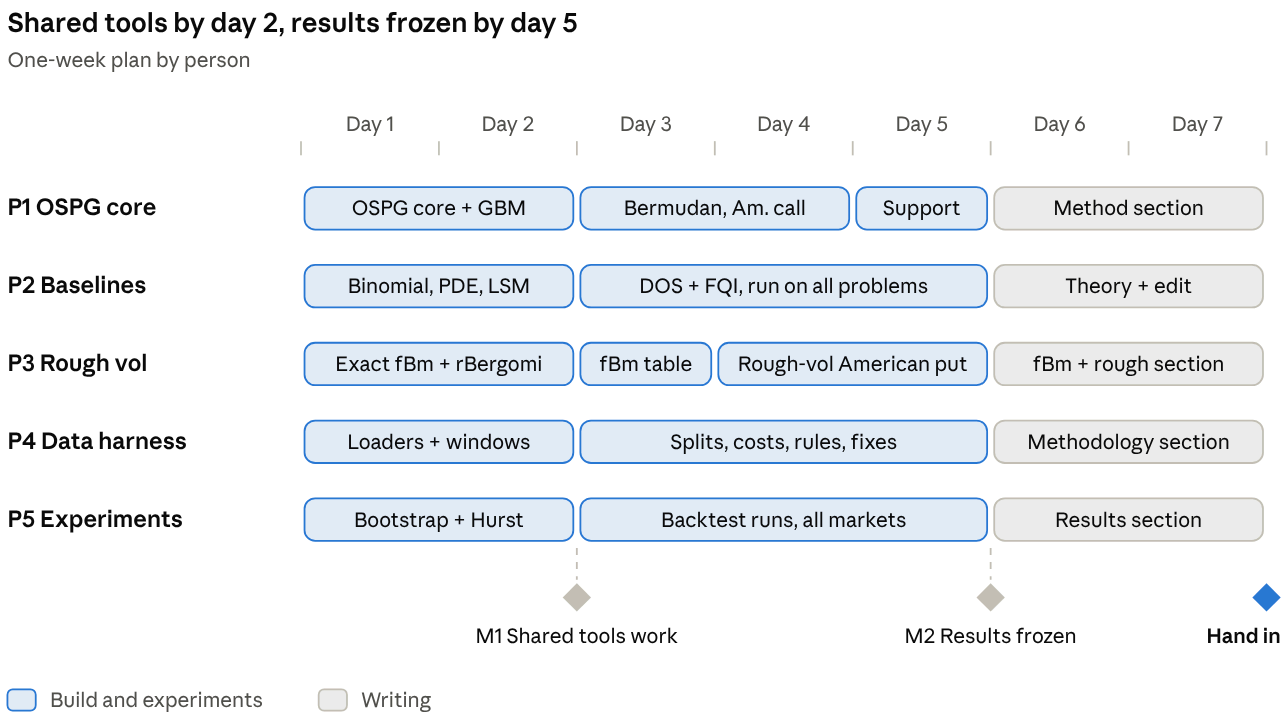
Every method exposes `fit(X_train, R_train, X_val, R_val)` and `stop(X) → τ` (one stopping index per path). Evaluation is always the mean of `R[i, τ_i]` on test paths.

```
ospg-project/
  ospg/          policy.py, loss.py, train.py          (P1)
  baselines/     lsm.py, dos.py, fqi.py, binomial.py,
                 pde.py                  (P2)
  simulators/    gbm.py (P1), fbm.py, rbergomi.py      (P3)
  data/          loaders/, clean.py, windows.py,
                 features.py, splits.py                (P4)
  backtest/      rules.py, costs.py, run.py            (P4)
  analysis/      bootstrap.py, hurst.py, regimes.py    (P5)
  experiments/   replication/, rough_vol/, backtest/   (configs + seeds)
  results/       tables and figures, one script each
  report/        LaTeX, one file per section
```

House rules: one GitHub repo, pull requests reviewed by one other person, fixed random seeds, every table regenerated by a script in `results/`.

Timeline

Days 1–2 build shared tools, days 3–5 run experiments, and days 6–7 are for writing.



7-day plan · 5 people · 2 milestones

M1 (end of day 2): the ospg package, simulators and data harness work end to end, which unblocks every experiment. M2 (end of day 5): results frozen; no new experiments after this point.

Report and presentation

Each person writes the section for their own work; P2 edits the whole report for one consistent notation.

Report section	Owner	Reviewer
Introduction and problem statement	P1	P2
Theory: Snell envelope, optional stopping, American put	P2	P3
Method: OSPG derivation and implementation	P1	P2
fBm and rough volatility: models and results	P3	P1
Data and backtest methodology	P4	P5
Backtest results and discussion	P5	P4

Report section	Owner	Reviewer
Conclusion and limitations	P2	all

Presentation: one slide block per person, in the same order as the report, with P2 assembling the deck.

Risks and fallbacks

The replication is low-risk; the rough-volatility and backtest workstreams carry most of the schedule risk, so each has a cut-down version.

Risk	Owner	Fallback
OSPG numbers don't match the paper	P1	Check the baseline is constant over time and the stopping-time probabilities sum to 1; report the gap honestly
Rough Bergomi simulator slow or unvalidated	P3	Use a simpler non-Markovian model (fBm-driven volatility) or fewer paths; cite reference prices only where matched
Not enough time for full replication grids	P1	Report a subset of the paper's settings and say so in the report
Crypto data blocked or patchy	P4	Switch exchange via ccxt; restrict to BTC and ETH
Backtest shows no gains	P5	Expected under near-martingale prices; frame as a test of optional stopping and lean on the risk-sensitive reward results
Look-ahead leakage	P4, P5	Time-based splits with gaps; flag any large mean-return gain as a leakage suspect first
Schedule slips	all	Cut in this order: FX and gold, the H sweep under rough vol, regime slices